INTERNSHIP PROJECT · VERACITIZ

Chronos AI

Watch Price Intelligence

An End-to-End Machine Learning Solution for Real-Time Price Prediction

1,488
Watch Records Analyzed

6
Regression Models Evaluated

71%
R^2 Score Achieved

Ready
Production-Ready Deployment

Machine Learning Internship Project · Veracitiz
https://github.com/Rudra2986/watch-price-prediction

01 · PROBLEM STATEMENT

The Valuation Challenge

Inconsistent Pricing

Watch prices vary significantly across brand, material, movement type, and design — making manual valuation unreliable and time-consuming.

No Scalable Solution

Existing approaches lack an automated, data-driven method to estimate prices at scale for e-commerce platforms.

Our Goal

Build an ML model that accurately predicts watch prices using historical data — from raw input to **real-time price output**.

This end-to-end pipeline transforms raw watch attributes into reliable price predictions, eliminating manual effort and inconsistency.

02 · PROJECT PIPELINE

Project Workflow

01

Raw Data

Collect the source watch records and attributes.

02

Data Cleaning

Remove errors, duplicates, and missing values.

03

EDA

Explore trends, patterns, and key relationships.

04

Feature Engineering

Transform raw inputs into model-ready signals.

05

Model Training

Train predictive models on prepared data.

06

Evaluation

Measure accuracy and compare model performance.

07

Hyperparameter Tuning

Optimize settings to improve prediction quality.

08

Deployment

Launch the final model for real-time use.

A streamlined end-to-end workflow that summarizes the full project from raw data to deployment, giving a fast, professional overview at a glance.

02 · TECHNOLOGY

Technology Stack

Display the complete technology stack used in the project with modern icons and clean visual cards:

Python

Programming Language

Pandas

Data Manipulation

NumPy

Numerical Computing

Scikit-Learn

ML Preprocessing

CatBoost

Gradient Boosting

Optuna

Hyperparameter Tuning

SHAP

Explainability

Streamlit

Web Deployment

GitHub

Version Control

A modern, production-grade tech stack combining data science, machine learning, and web deployment tools.

03· DATASET

Dataset Overview

The **WatchVine E-Commerce Dataset** was scraped and cleaned to support production-grade watch price modeling.

3000+
Raw Columns
Initially scraped, later reduced

1,488
Records
Final cleaned dataset

25+
Features
Meaningful input variables

15
Brands
Distinct watch brands represented

Main feature categories

1	2
Brand Brand identity and positioning	**Material** Case and strap materials
3	4
Watch Type Analog, digital, or hybrid formats	**Category** Product classification
5	6
Dial Color Visual style of the watch face	**Strap Color** Primary strap appearance
7	8
Movement Type Quartz, automatic, or other mechanisms	**Case Diameter** Size dimension in millimeters
9	10
Water Resistance Durability and usage rating	**Is Automatic** Binary indicator for automatic movement

The dataset combines brand, design, and technical specifications to create a strong feature set for pricing analysis and predictive modeling.

04 · DATA PREPARATION

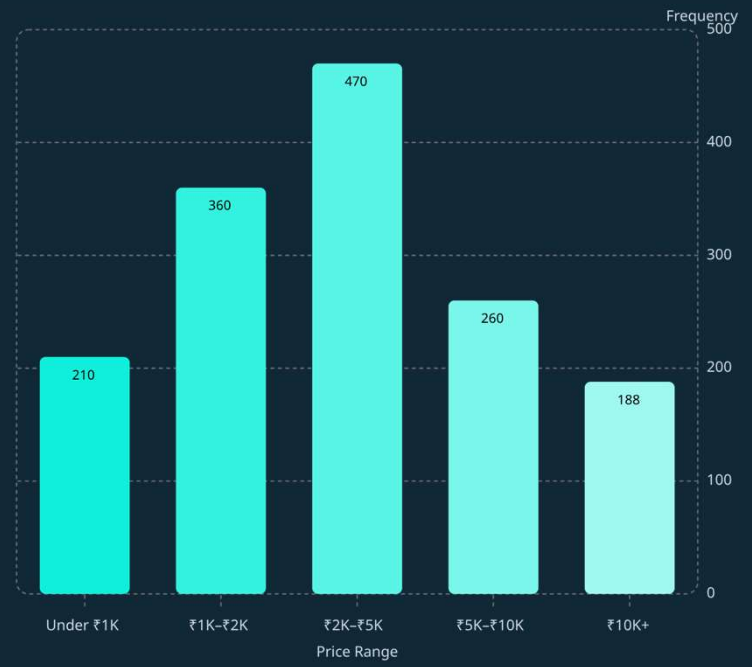
Data Cleaning & Exploratory Data Analysis

Cleaning Pipeline

- Removed 3000+ irrelevant scraped columns
- Converted price strings to numeric INR values
- Handled missing values & standardized text fields
- Removed leakage-prone columns

These steps produced a cleaner, model-ready feature set with consistent formatting and reduced noise.

Price Distribution



The distribution is concentrated in the mid-range, with fewer watches at the premium end, indicating a right-skewed pricing pattern.

05 · FEATURE ENGINEERING

Feature Engineering

Feature engineering transformed raw watch data into structured inputs that improved model performance and made patterns easier for the model to learn.

Binary Features

Is Automatic and **Is Luxury Brand** capture key yes/no signals that are easy for the model to interpret.

Categorical Encoding

Brand, **Material**, and **Watch Type** were encoded to convert text categories into usable numeric features.

Numerical Scaling

Case Diameter and **Water Resistance** were scaled to keep numeric features on comparable ranges.

Feature Selection

Redundant and leakage-prone columns were removed to keep only the most informative variables for training.

Encoding Strategy

- One-hot encoding for low-cardinality categories
- Target encoding for higher-cardinality features
- Ordinal handling for ordered numeric-like attributes

Impact

The final feature set was cleaner, more consistent, and more predictive — helping the model learn stronger relationships and achieve better performance.

06 · MODEL DEVELOPMENT

Machine Learning Model Development

Six regression algorithms were trained and compared using **5-Fold Cross Validation** to identify the best-performing model.

Linear Regression
Baseline model that fits a straight-line relationship between inputs and the target.

Ridge Regression
Regularized linear model that reduces the impact of correlated features.

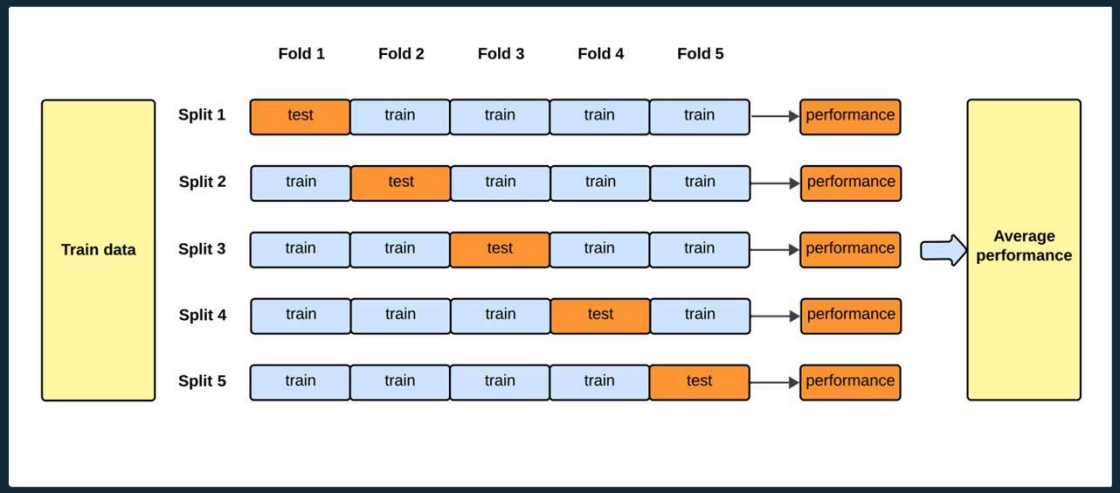
Random Forest
Ensemble of trees that captures non-linear patterns and interactions.

XGBoost
Gradient boosting method that sequentially improves accuracy.

LightGBM
Fast boosting framework designed for efficient training on larger datasets.

CatBoost
Boosting model that handles categorical features with minimal preprocessing.

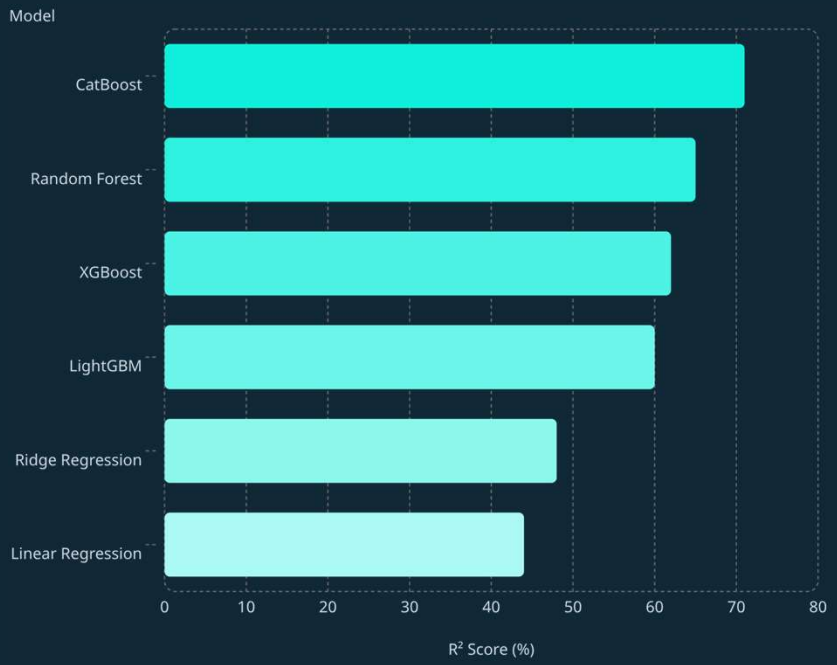
5-Fold Cross Validation



This process provides a more reliable estimate of real-world performance by testing each model across multiple data splits. **Final performance = Average of 5 validation scores**.

06 · EVALUATION

Model Evaluation & Selection



Selected as Final Model: Highest R^2 score ( ) with superior handling of categorical features and consistent cross-validation performance.

Why CatBoost?

Best Fit

Highest variance explained across all six models.

Categorical Strength

Native handling of categorical features with minimal preprocessing.

Consistent Validation

Strong and stable performance across 5-Fold Cross Validation.

Model Comparison Table

Model Name	R^2 Score
Linear Regression	
Ridge	
Random Forest	
XGBoost	
LightGBM	
CatBoost (Winner)	

07 · OPTIMIZATION

Hyperparameter Tuning & Explainable AI

Optuna — 100 Trials

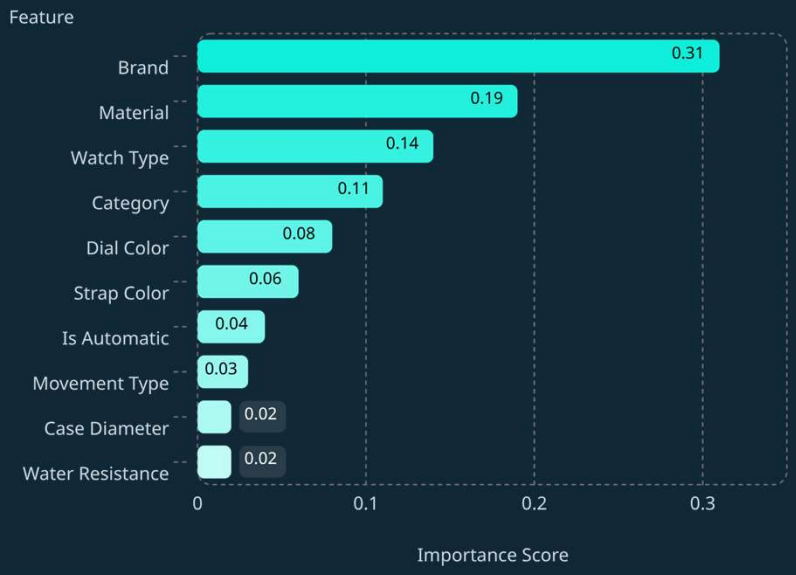
Automated Bayesian optimization to find the best CatBoost setup.

Iterations	Depth
540	5

Learning Rate
0.0465

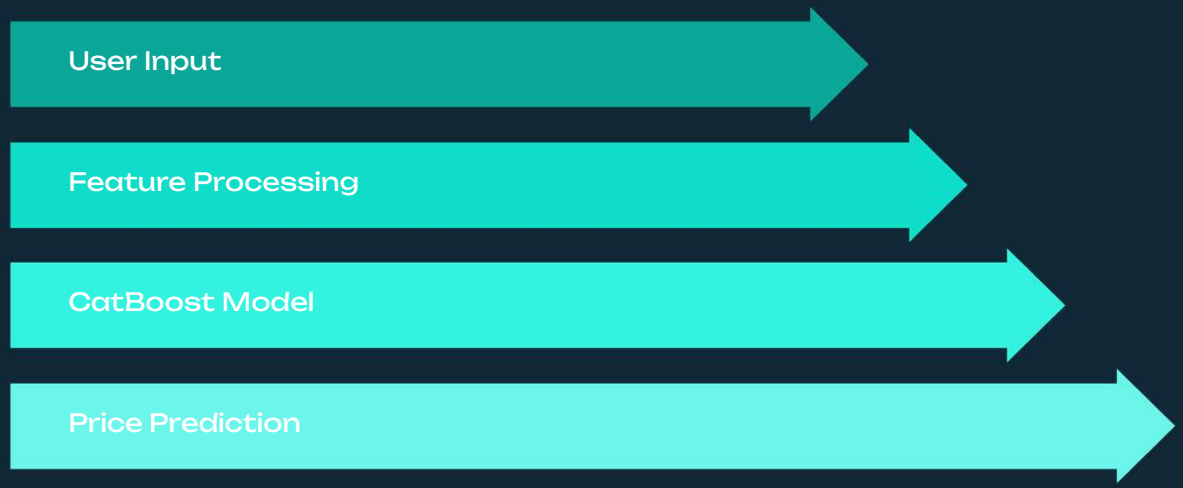
SHAP Explainability

SHAP values reveal which features most influence price predictions, enabling model interpretability and trust.



08 · DEPLOYMENT

Deployment Architecture



From user input to price output in seconds — the Chronos AI pipeline delivers production-ready predictions through a clean, intuitive interface.

Architecture Overview

User Input
Watch attributes collected via interactive form

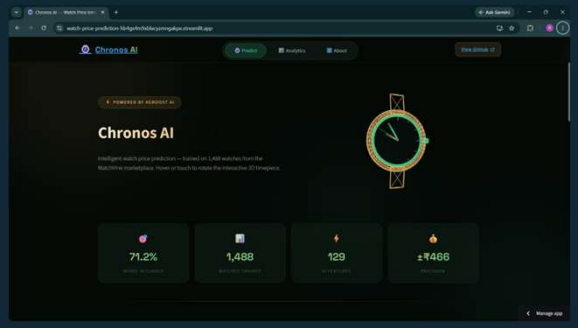
Feature Processing
Raw inputs transformed into model-ready features

CatBoost Model
Trained regression model making predictions

Price Prediction
Real-time output with confidence scores

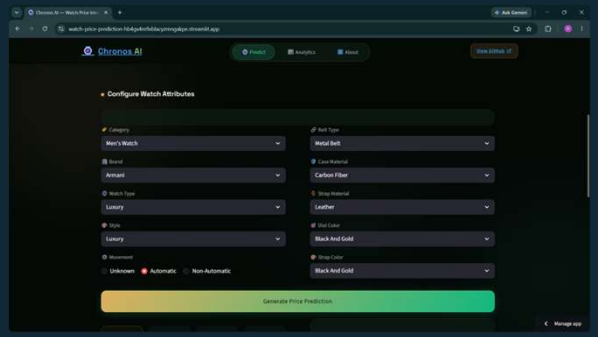
09 · LIVE APPLICATION

Chronos AI — Live Application



Home Page

App overview and navigation



Prediction Page

Enter watch attributes

Results Page

Price output and explanations

Chronos AI is deployed as a production-ready Streamlit web application, enabling real-time watch price predictions with explainable AI insights.

10 · RESULTS & ROADMAP

Results, Challenges & Future Scope

71%

R^2 Score

Final CatBoost model performance

₹466

RMSE

Root Mean Squared Error

₹294

MAE

Mean Absolute Error

0.07%

Brand Unknown

— critical fix

Business Impact

- Automated watch valuation
- Reduced manual price estimation effort
- Scalable for e-commerce platforms
- Real-time prediction capability
- Production-ready deployment

🏆 Challenge Solved

Brand Sparsity Problem:

87.5% → 0.07%

Unknown brands reduced through intelligent feature engineering, driving significant model performance gains.

🔭 Future Scope

- Larger, more diverse datasets
- Image-based watch analysis (CNN/ViT)
- Text embeddings from product descriptions
- Automated retraining pipeline
- Cloud deployment (AWS / GCP)

> Successfully built an end-to-end Machine Learning solution — from raw scraped data to production-ready deployment. **Chronos AI is live.**

11 · KEY LEARNINGS

Key Learnings & Achievements

This project provided comprehensive experience across the machine learning lifecycle, highlighting mastery in:

- End-to-End Machine Learning Pipeline
- Data Cleaning & EDA
- Feature Engineering
- Regression Modeling

Achievement Badges

- End-to-End ML Pipeline
- Explainable AI
- Hyperparameter Optimization
- Production Deployment

This project demonstrates mastery of the complete machine learning lifecycle, from data acquisition to production deployment.

Thank You

Questions & Discussion Welcome

Rudra Patel · Machine Learning Intern · Veracitiz

12 · PROJECT LINKS

Project Links & Resources

GitHub Repository

Complete source code, notebooks, and documentation

GitHub - Rudra2986/watch-price-prediction: Watch price prediction model using ML

Live Application

Try Chronos AI in action

Chronos AI — Watch Price Intelligence

GitHub Repository QR Code

Live Application QR Code

Scan to access the project or visit the links above